Customer Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

Dataset Overview

3,900

Purchase Records

Transactional data analyzed

18

Data Columns

Customer and purchase details

37

Missing Values

Review ratings imputed

Data Preparation Process

01

Data Loading

Imported using pandas

02

Exploration

Checked structure and statistics

03

Cleaning

Handled missing values

04

Standardization

Renamed columns to snake case

05

Feature Engineering

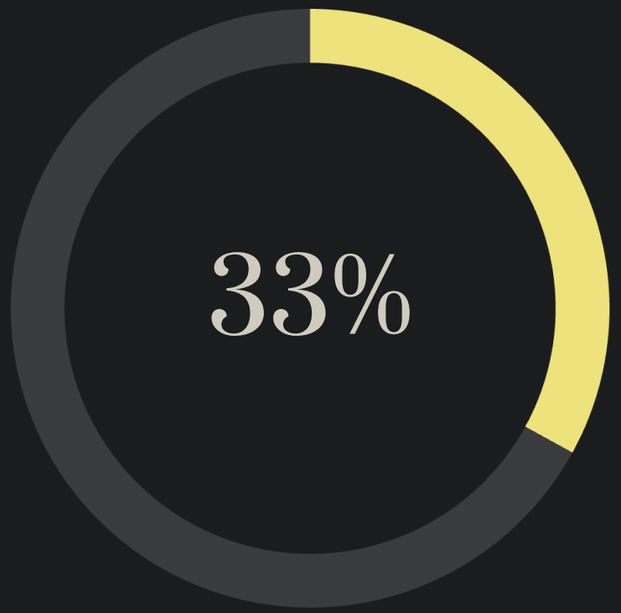
Created age groups and frequency metrics

06

Database Integration

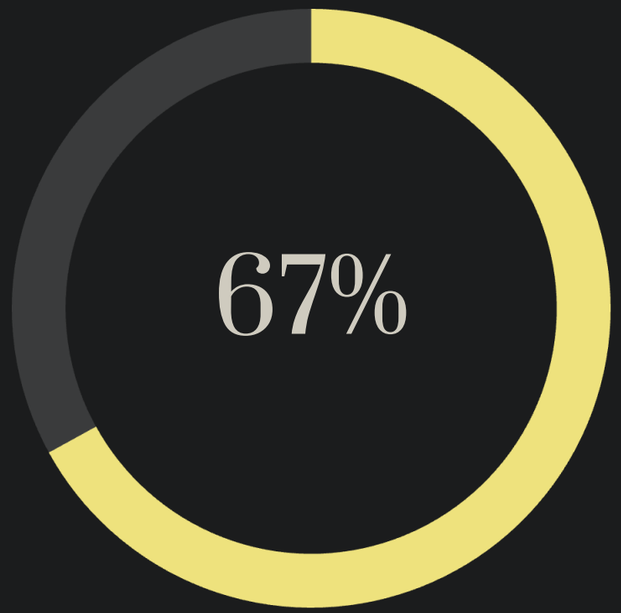
Loaded into PostgreSQL

Revenue by Gender



Female Customers

$75,191 revenue



Male Customers

$157,890 revenue

Product Performance Insights

Top Rated

Gloves, Sandals, Boots lead ratings

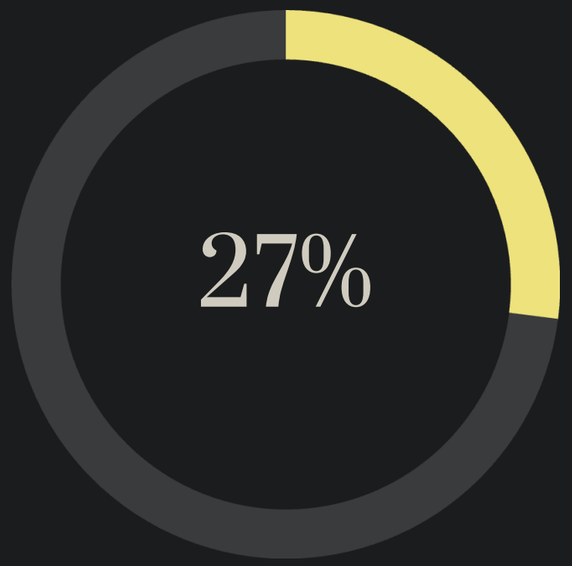
Best Sellers

Blouse, Pants, Jewelry most purchased

Revenue Leader

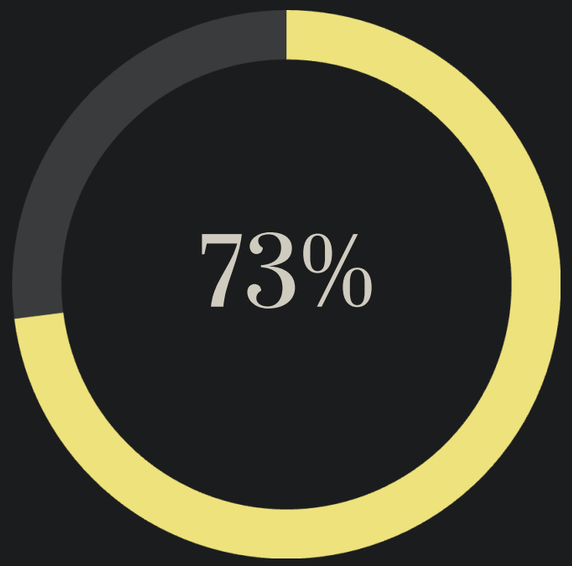
Clothing: $104K revenue

Subscription Analysis



Subscribers

1,053 customers



Non-Subscribers

2,847 customers

Average Spend

Subscribers: $59.49
Non-Subscribers: $59.87

Total Revenue

Subscribers: $62,645
Non-Subscribers: $170,436

Shipping & Discount Behavior

Express Shipping

Average purchase: $60.48

Standard Shipping

Average purchase: $58.46

High-Spending Discount Users

839 customers used discounts above average

Age Group Analysis

27%

Young Adult

$62K revenue, 2.4M sales

25%

Middle-aged

$59K revenue, 1.9M sales

24%

Adult

$56K revenue, 1.8M sales

24%

Senior

$56K revenue, 1.8M sales

Key Findings

Repeat Buyers Subscribe

958 repeat buyers subscribed vs. 2,518 non-subscribers

Category Performance

Clothing dominates revenue ($104K) and sales (3.4M)

Age Group Parity

Revenue evenly distributed across all age groups

Strategic Recommendations

Boost Subscriptions
Promote exclusive benefits

Loyalty Programs
Reward repeat buyers

Review Discounts
Balance sales and margins

Target Marketing
Focus on high-revenue groups